SAMPLE

Topological Quantum Matter

New States of Matter

Aira Astralis

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Introduction

Physics spent three centuries classifying matter by what it looks like: gas, liquid, solid; crystal and glass; iron's magnetism and copper's shine; the phases drawn on a map whose axes were density, order, and symmetry. The map worked because the distinctions were local: a solid differs from a liquid in how its atoms sit next to their neighbors, a magnet differs from ordinary iron in the local alignment of its atoms' spins, and every phase transition the classical world knows is a rearrangement of local patterns. The twentieth century's last decades added a second map, drawn with a different mathematics, and the second map's regions do not differ in any local pattern at all. They differ in their global architecture, in the way the whole material's quantum wave function is connected up across its extent, in quantities that no local measurement can see and no local disturbance can change. The mathematics of such global properties is topology, the study of what survives smooth deformation, and the materials whose behavior is governed by it are topological quantum matter, the subject of this book: insulators that conduct only at their edges, particles that are not particles but fractions of one, states that remember their history in braids, and a new classification of matter in which the deepest properties are not what the material does, but what it cannot stop doing.

The reader should be told, at the start, why the trade cares, because the reason is rare in physics: robustness. Every ordinary material property is an accident of preparation; purity, temperature, and disorder push it around, and the engineer's career is spent fighting the drift. A topological property is different in kind: it is a counting number, an integer or a fraction that cannot change continuously, and to destroy it one must close a gap, break the material's bulk, or drive a phase transition outright. The consequences are the book's recurring wonders: electrical resistances that fix themselves to fundamental constants with a precision the metrologists use to define the units of resistance, currents that flow along a material's edge because the bulk forbids them and the mathematics requires them, and quantum information's oldest enemy, decoherence, promised a genuine defense for once, in materials whose very mistakes are topologically checked. The trade found, in the second map, properties as reliable as arithmetic, and the discovery is quietly rearranging the deep physics of materials and the engineering hopes of quantum computing alike.

The book tells the story whole, as the series does: the mathematics of coffee cups and doughnuts first, then the experiments that forced it on the trade, the quantum Hall effect's impossible exactness and the fractions it hid; the insulators that conduct at their skins, the particles that are braids, the superconductors that host half-electrons; the Nobel-labeled chronology and the honest status of the contested claims; the machines built to use the properties, and the questions still open, including the largest one, what order itself is. The reader needs no mathematics beyond the book's prose, and the honest account of what is established, what is engineered hope, and what remains a wager, is the spine of the telling, as it is throughout this series.

A note on the book's honesty, the series' standing practice, stated here because this subject needs it more than most. Topological quantum matter is an active field with a hype cycle: the words topological and quantum appear in funding documents, press releases, and product pitches, sometimes deservedly and sometimes decoratively, and the chapters ahead sort the claims as the trade itself does. Established physics is labeled established; engineering hopes are labeled hoped; contested signatures are labeled contested, with the field's own standards stated; and the reader who wants to know what is real can always find the ledger chapter, and the book's every claim can be checked against it. The exactness the subject is famous for is worth this care: the topological integers really are the most reliable properties in materials science, and they deserve a book that keeps the rest of the prose as honest as the integers.

Chapter 1: The Discovery: When Geometry Changed the Trade

The subject's official birth is dated by the trade to a single October morning in 1985, but the story properly begins with a curiosity that had been annoying condensed-matter physics for decades: the strange behavior of thin things. The physics of the bulk, the three-dimensional world of ordinary materials, was largely solved business by mid-century, its phases classified by symmetry and its transitions catalogued; but the physics of very thin films and very cold layers kept producing anomalies the classification could not digest. In the 1970s, the study of two-dimensional superfluids and magnets found transitions that the standard machinery said could not happen, flat-world phase changes with no symmetry anywhere breaking, and the young physicists who explained them, Michael Kosterlitz and David Thouless, found the answer in a topological object, the vortex, and a topological mechanism, the unbinding of vortex pairs: the flat world's transitions were not about symmetry at all but about topology, the first crack in the classical map.

The full break came with the quantum Hall effect, and the reader should feel its scandal. In 1980, Klaus von Klitzing, measuring the resistance of a cold two-dimensional electron layer in a strong magnetic field, found the resistance plateau, as he turned the field's knob, at values locked to integer fractions of a fixed combination of fundamental constants, and the locking was exact to the instruments' last digit, the resistance reproducing to parts in a billion regardless of the sample's purity, its geometry, its temperature, everything a physicist expects to matter. Exactness of that kind does not happen in messy solids by accident; it had, in the entire prior history of condensed matter, essentially never happened, and Thouless and the community that cracked the plateau's meaning found the explanation in a topological integer, a winding number of the electrons' quantum wave function, a global count that no local disorder could touch. The two discoveries, the flat-world transitions and the impossible plateau, converged on one mathematics, and the 2016 Nobel Prize that crowned the three founders, Thouless, Kosterlitz, and Duncan Haldane, honored the convergence: the recognition that a large part of matter's deepest behavior is topological, and that the trade had been given a second map.

The discovery's method deserves its paragraph, because it is the reverse of the usual one. The great discoveries of physics are usually predicted and then found; the topological ones were found and then understood, the instruments leading the mathematics by years in every case: the plateau was measured before its integer was understood, the fractional plateaus were measured before their anyons were theorized, and the topological insulators were measured before the band-structure community finished its accounting. The reader should register the pattern as the subject's signature: this is a field where the materials keep arriving ahead of the explanations, which has made it, for four decades, the happiest hunting ground in condensed matter, an experimentalist's subject wearing a theorist's clothing.

The prize's citation deserves quoting in the book's spirit, because the Nobel committee's words captured the subject's essence better than most popular accounts: the three laureates were honored for the discovery of topological phase transitions and topological phases of matter, the first time the prize's citation used the word topology, and the trade's response was a generation's worth of pride, the founders of the flat-world physics and the band-structure counting crowned together, decades after the work. The reader should note the prize's timing as the subject's own measure: the transitions were explained in the early 1970s, the band counting in the early 1980s, and the crown came in 2016, after the topological insulators had made the subject unignorable, the pattern being the one this book has shown throughout, the mathematics waiting for the materials to make it famous.